

FigmaXR — IP2 Evaluation

1) Objective & validation metrics

This evaluation focused on testing how effectively FigmaXR supports UI testing inside a virtual environment, and whether users can interact with devices and environmental controls in intuitive and meaningful ways. The prototype was tested using Meta Quest 2/3 headsets in a single VR office scene with three devices, adjustable lighting, an audio toggle, and room navigation.

The study followed a structured, task-based approach supported by observation and short interviews. Success was defined through task completion rate, interaction accuracy, time efficiency, and subjective comfort levels. Participants were asked to pick up devices, adjust the light, toggle background audio, and move between rooms. The evaluation also aimed to see whether users could understand the purpose of these interactions—something that seemed less clear in the previous prototype.

2) Results

All five participants completed the four core tasks, but the way they interacted revealed several recurring issues. Everyone managed to grab the blue device, though several needed multiple attempts or asked which button to press. All adjusted the brightness, but four said the change wasn't obvious or only lit one part of the room. Everyone turned on the music, but two initially couldn't hear it, suggesting sound delay or low volume. Everyone reached the target area, though most had trouble locating the entrance. Two preferred teleportation to joystick movement because it felt faster and less disorienting.

Common feedback included that the scene felt “too plain” or “unfinished,” and the purpose of the tasks was unclear. One participant said, “I didn't realise this was supposed to be a café,” while another commented that the interactions “felt like buttons on a wall rather than XR experiences.” Despite these issues, all users said the system worked smoothly overall and could imagine its potential once the interactions felt more spatial and less abstract.

3) Analysis / insights

The tests highlighted three main areas that need attention.

First, spatial understanding and orientation. Although the navigation system functioned correctly, users often got lost or weren't sure what the rooms represented. The entrances weren't visually distinct, which made the environment feel repetitive. The preference for teleportation over walking suggests that physical realism can sometimes harm usability when orientation cues are weak.

Second, interaction realism and feedback. Most users enjoyed being able to pick things up and control lighting, but several mentioned that the experience felt “flat.” They expected interactions that used the body—like squeezing light, pulling a lever, or moving sound in 3D space—instead of sliders and toggles. The current design behaves like a 2D interface transplanted into VR, missing the opportunity to use spatial affordances.

Third, conceptual clarity. The most consistent confusion was about why they were doing these tasks. Without an introductory cue or narrative, participants didn't connect the interactions to UI testing. This gap between system function and conceptual meaning reduced engagement and limited the quality of their feedback.

4) Evaluation of aims

The evaluation partially met its original aims.

Functional success: All core mechanics worked reliably, validating the prototype's stability.

Accuracy and feedback: Technically correct, but visually underwhelming—users couldn't always tell if their actions had an effect.

User comfort: Generally positive, though joystick navigation caused mild discomfort for one participant.

Comprehension and engagement: Not achieved. Participants could complete tasks but did not grasp the broader purpose of the system.

Perceived usefulness: Mixed. Users saw potential but felt the prototype was still too conceptual to demonstrate clear benefits over standard testing methods.

5) Concept iteration

Based on these insights, I plan several concrete changes for the next iteration. Navigation will be made more legible by adding visual door frames, colour-coded paths, and light gradients between rooms. Walking will remain an option, but teleportation will be added for comfort and accessibility. Lighting control will be redesigned as a "light orb" that users can grab and resize, affecting the entire space with more noticeable feedback. The audio interaction will be turned into a physical metaphor such as a record or speaker that users can twist or pull, supported by haptic feedback. Object selection will be improved with raycasting and vibration cues, so users can grab items confidently from a distance. A short tutorial or text panel will explain that the system is meant for testing UI readability under different environments. Finally, each room will have its own visual and acoustic identity, such as a café, studio, or office, to make environmental changes meaningful.

6) Reflection & next test

This round of testing taught me that working interactions are not necessarily understandable ones. Technically, the prototype functioned without major issues, but users struggled to understand the purpose behind what they were doing. That gap is just as important to fix as any bug.

It also reminded me that in VR, spatial design is communication. When environments lack cues, users feel lost even if the system works perfectly. And when interactions mimic 2D interfaces, they lose the expressive power that makes XR compelling.

Combining timing data with short interviews was effective. The numbers confirmed that users could complete tasks efficiently, while the interviews revealed why they felt unsure or disconnected. In future studies, I plan to add a short training segment and a clearer structure for verbal feedback, possibly using the IUQ scale to link subjective impressions with task data.

Overall, this evaluation showed that FigmaXR has solid technical foundations but needs stronger environmental storytelling and embodied interaction to truly deliver on its goal—testing interface usability in immersive, meaningful ways.

Appendix

See *EvaluationResult2_Rawdata.pdf* for the complete dataset.